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Editors
Automated Software Engineering

Dear Editors,

PRE-SUBMISSION DRAFT — DO NOT UPLOAD YET

Re: Original Paper submission

“From Traces to Witnessed Contracts: Incremental Precondition–Effect Compilation for LLM Agent Skills”

Please consider our manuscript for publication as an Original Paper in *Automated Software Engineering*. The paper addresses a practical maintenance problem created by executable LLM-agent skills: a plausible revision may retain its final goal action while silently breaking navigation, possession, or transformation dependencies.

We introduce Incremental Witnessed Contract Compilation (IWCC), a pre-execution validator that compiles successful native traces into deterministic, provenance-carrying contracts over objective roles, milestones, precedence, navigation, and producer–consumer dataflow. Each learned clause retains its successful witnesses, and each rejection identifies a violated clause and action position.

The evaluation separates exposed development evidence from a frozen, zero-overlap confirmation. Across 60 ALFWorld and ScienceWorld tasks and 300 candidate programs, IWCC reaches 88.12% balanced accuracy, compared with 77.48% for the predeclared strongest comparator. The 10.64-point gain has a task-stratified 95% confidence interval of 8.88–12.50 points and a paired task permutation p -value of 1.0×10^{-5} . IWCC accepts all 98 healthy or behaviorally equivalent controls and runs at 0.487 ms median latency on one CPU process.

The work fits the journal’s scope by treating reusable agent skills as evolving software artifacts and combining specification mining, software validation, provenance, and release-time defect screening. The claim is deliberately bounded: IWCC detects modeled structural defects but does not certify semantic success or replace native execution.

Source code, data, frozen protocols, analysis scripts, and the reproducibility bundle are publicly available at <https://github.com/ls680/iwcc-agent-skill-contracts> under Apache-2.0.

Before this letter is uploaded, the versioned archival DOI must be inserted, and both authors must confirm originality, exclusive submission, author order, and approval of the submitted version. The authors declare no competing interests and no specific funding. Generative-AI assistance is disclosed in the manuscript.

Sincerely,

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